LARGE LANGUAGE MODELS IN HUMAN-MACHINE INTERACTION

Ömer Emre Mutlu

emre.mutlu@rwth-aachen.de

Abstract

Large language models have recently emerged as a powerful interface paradigm for human machine interaction, particularly in safety critical domains where speed, reliability, and cognitive support are essential. This work presents a structured literature review of how large language models are integrated into safety critical human machine interaction systems across text based, audio based, and visual language modalities. Rather than focusing on standalone model capabilities, the review examines the architectural roles assigned to large language models within constrained interaction pipelines, including retrieval augmented generation, tool calling, agentic reasoning, speech to speech processing, and vision language integration. Representative case studies from aerospace operations, emergency and environmental response, and industrial settings are discussed. Across these domains, the findings indicate that hybrid architectures combining large language model based high level reasoning with deterministic low level execution and human in the loop control provide the most reliable and usable solutions. While challenges related to bias, latency, multimodal integration, and robustness remain, the reviewed evidence suggests that carefully constrained multimodal large language model based interfaces can reduce operator workload and enhance decision support without compromising safety.

Keywords: Large Language Models, Human Machine Interaction, Visual Language Models, Multimodal Systems, Speech Based Interfaces, Retrieval Augmented Generation, Tool Calling, Agentic Artificial Intelligence, Safety Critical Systems, Human in the Loop Systems

CONTENTS

1 Introduction 3
2 Background 3
2.1 Text-Based LLM Interfaces 4
2.2 Audio Based Interfaces 6
2.3 Vision-Language Models 8
3 Domain Case Studies 9
3.1 Aerospace and Remote Science 9
3.2 Emergency and Environmental Response 14
3.3 Industrial Operations 16
3.4 Discussion 17
4 Conclusion 19

Disclaimer: Artificial intelligence tools were used strictly for grammar correction and minor editorial polishing; all analysis, arguments, and original content remain solely the work of the author.

1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human–machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens [RSS+25]. In regulated application domains specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder–decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field [CYJ+25]. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities [CFD24].

While a growing body of literature examines large language models from the perspective of architecture design and task performance, comparatively less attention has been given to their role within safety critical human machine interaction systems. In these settings, large language models are rarely deployed as autonomous decision makers. Instead, they are embedded within constrained system architectures that emphasize human supervision, deterministic execution, and compliance with safety regulations. As a result, understanding how large language models mediate interaction, support operator decision making, and coordinate complex tasks is essential.

This paper presents a structured literature review of large language model based human machine interaction in safety critical environments. The focus is placed on the interaction roles, architectural patterns, and degrees of autonomy assigned to large language models across different domains, rather than on standalone model capabilities or performance benchmarks. The review synthesizes how text based, audio based, vision based, and multimodal interaction mechanisms are combined within hybrid architectures to enhance usability and coordination while preserving safety and human authority. First, the architectural foundations of large language model based interaction systems are introduced. This is followed by a discussion of representative case studies from selected safety critical domains.

2 Background

Before proceeding with the analysis of domain-specific work, a brief introduction to the involved technologies is necessary to provide the reader with sufficient background and contextual understanding. Accordingly, this section presents a concise overview of the relevant technologies.

In a human–machine interaction (HMI) system, the human can be regarded as a component equipped with four primary input–output channels: vision, hearing, touch, and motor activity (movement) [DFAB04]. Among these channels, vision and hearing are most commonly addressed in contemporary AI-driven interaction systems and are therefore particularly relevant in the context of large language model (LLM) based interfaces. For the visual modality, current approaches primarily involve text-based language models and vision–language models that process textual input alongside visual information. For the auditory modality, interaction is typically realized through audio-based processing pipelines that combine speech recognition and speech synthesis with language models. The subsequent subsections further examine these modalities in more detail.

2.1 Text-Based LLM Interfaces

Since the introduction of the Transformer architecture in "Attention Is All You Need", text-based content generation and processing have become the central foundation of modern natural language processing (NLP).[VSP+17] In this context, large language models are a family of neural language models trained on vast amounts of textual data and capable of performing a wide range of natural language processing tasks, such as generating context-aware responses based on user-provided text prompts.[BMR+20] To utilize this family of models as a user interface, the model must be capable of understanding domain-specific context and of acting upon user prompts, rather than merely generating text responses. Several methods exist to enable this, a brief summary of these approaches is provided below.

Fine Tuning

Fine-tuning is the process of further training a pre-trained model on task- or domain-specific data by updating its existing parameters or, in some approaches, by introducing additional trainable parameters.[Mos23] By further training the model on a domain-specific dataset, for example technical data on extraterrestrial habitats, a base model such as GPT-5 can acquire contextual knowledge of the application domain and generate its responses accordingly.

However, a significant downside of this approach is the increased risk of model hallucinations. Hallucinations refer to the phenomenon in which a model generates incorrect or fabricated information in response to a given prompt.[GYA+24] Such behavior can pose a significant risk in safety-critical environments.

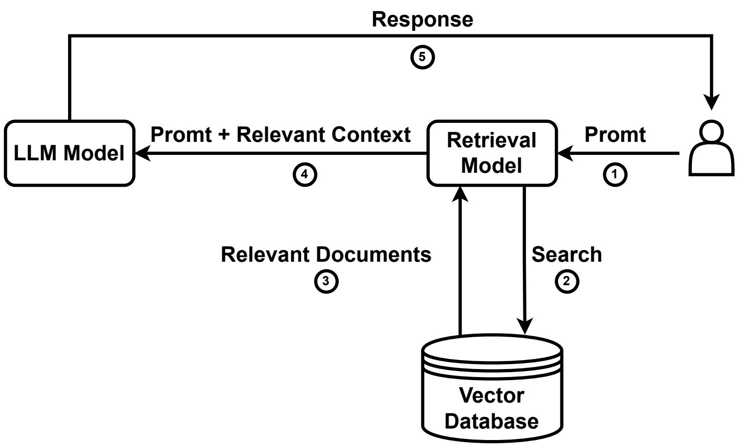


Figure 1: High-level RAG workflow that injects retrieved documents into the prompt instead of altering model weights.

Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) introduces relevant external information to the language model without modifying the model's internal parameters.

As shown in Figure 1, the RAG workflow begins with a user prompt; however, instead of passing the prompt directly to the LLM, a similarity search is first performed over context-relevant information stored as embeddings in a vector database. The retrieved context is then provided together with the prompt to the LLM for grounded response generation.[BMR+20] Recent studies have shown this to be a more reliable method for context retrieval compared to fine-tuning.[OBME24]

However, the risk of hallucinations remains non-zero even with RAG; nevertheless, with well-designed context selection and retrieval strategies, this probability is significantly reduced, as demonstrated in numerous studies [JLF+24].

Tool Use / Function Calling

Tool calling, also referred to as function calling, is the ability of a large language model to invoke external functions by interpreting user context and generating structured Application Programming Interface (API) calls with the relevant parameters extracted from user prompts, as introduced by Schick et al. [SDYD+23].

In a practical setting, this means that the model generates structured tool calls, which are then used by a service layer to invoke HTTP requests to downstream services for task execution, as illustrated in Figure 2. By utilizing this method, the model can interact with external systems to trigger the execution of real-world functions.

In the context of human–machine interaction, this implies that a user can issue natural-language prompts to control robots and machines to execute a wide variety of tasks, as demonstrated by Vemprala et al. [VBBK23].

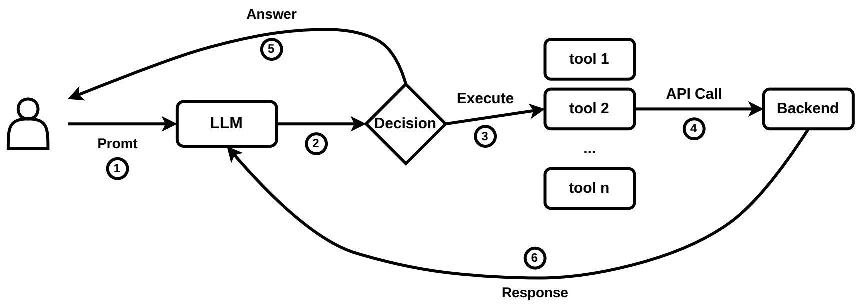


Figure 2: Conceptual workflow of tool (function) calling in a large language model.

Agentic Artificial Intelligence

What is commonly referred to as agentic AI is the combination of previously discussed methods to enable semi-autonomous decision-making systems. An AI agent typically integrates a large language model for reasoning and planning with tool-calling capabilities and memory in order to execute tasks without constant human supervision, requesting user input only when necessary to adjust its actions [YZY+23].

Such agents are currently most commonly used in programming-related tasks; a prominent example of an agentic AI system in this domain is OpenAI's Codex-based coding agent.

With regard to the use of such systems in human–machine interaction, Kim et al. note that the application of agentic systems in this context remains underexplored. However, their user study reveals that current agentic systems are still prone to failure, which caused usability issues for some participants [KLM24]. The agents in their experiment exhibited persistent hallucinations, leading some participants to doubt their own knowledge. Additionally, in cases where a physical robot embodiment was used to represent the agent, participants reported an uncanny-valley-like sense of unease during task execution. Nevertheless, the field and the underlying technology are still at an early stage of development, and user experience is expected to improve over time. The application of such systems in domains such as deep-space exploration remains a particularly promising research direction, as discussed by Heinicke et al. [FSG+21]. However, in their current state, the suitability of such systems for safety-critical applications remains uncertain.

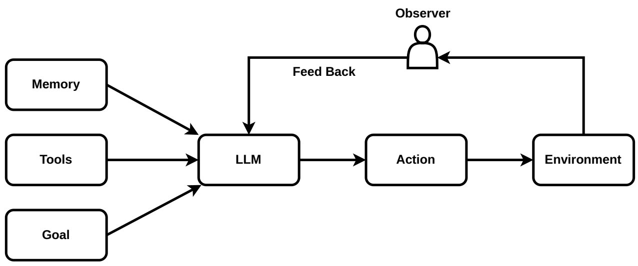


Figure 3: High-level agentic AI stack combining planning, memory, tool use, and execution loops.

2.2 Audio Based Interfaces

A critical component of human–machine interaction is speech, which represents one of the most natural and powerful means of human communication. In many HMI application domains, the ability to translate spoken user input into corresponding machine actions is of particular importance. Accordingly, this section discusses two methods that are currently employed to enable speech-driven interaction in such systems.

Speech-to-Text, LLM and Text-to-Speech

As highlighted by Haeb-Umbach et al. and other previous work, until recently this architecture represented the primary approach for providing voice support in interactive human–machine interaction (HMI) applications [HUWN+]. The overall pipeline typically consists of a speech detection and preprocessing layer, which forwards the audio signal to a speech-to-text (STT) model. The spoken input is thereby converted into text and subsequently processed by a large language model (LLM), which performs the reasoning and retrieval operations introduced in Section 2, such as retrieval-augmented generation (RAG). The LLM then produces a textual response, which is finally converted back into speech and presented to the user via a text-to-speech (TTS) system.

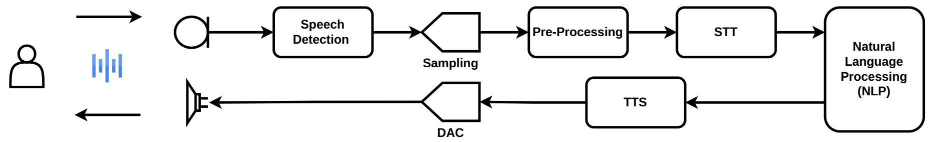
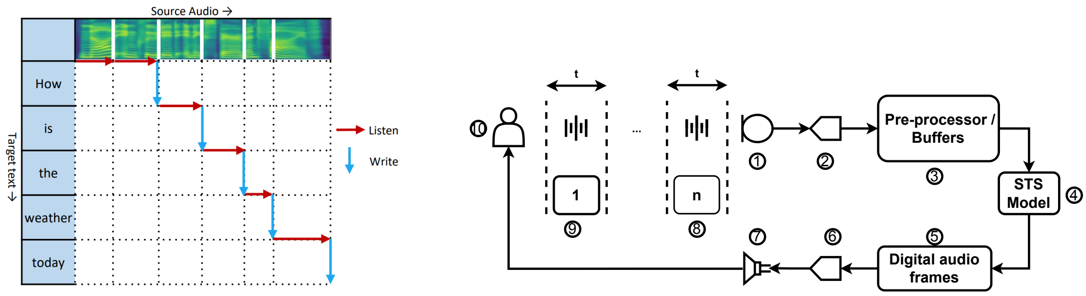


Figure 4: Signal chain for speech-to-text ingestion, LLM reasoning, and text-to-speech playback.

However, this pipeline also exhibits notable drawbacks. The sequential use of speech-to-text (STT), large language model (LLM) processing, and text-to-speech (TTS) synthesis can introduce perceptible latency. To mitigate this effect, some systems employ interaction design strategies such as playing short default utterances (e.g., “Let me think about that…”) while the main response is being generated. Nevertheless, empirical user studies indicate that such waiting times can still negatively affect the naturalness and fluidity of human–machine interaction, particularly in conversational settings [FSG+21, KLM24].

On the positive side, such a configuration allows spoken input to be converted into textual form, which can facilitate integration with retrieval-augmented generation (RAG) systems and thereby support more efficient retrieval of contextually relevant information.

Speech-to-Speech LLMs



(a) Streaming wait-k policy alternating between listening to additional source frames and emitting target tokens in real time [TLR+20].

(b) Streaming signal chain diagram with numbered legend.

Legend: 1) Microphone interface capturing the operator’s raw speech
2) Low-latency Analog to Digital Converter
3) Pre-processing buffers that normalize, filter, and batch frames
4) Streaming STS model performing NLP task
5) Generated digital audio frames for the target utterance
6) Vocoder/decoder producing analog waveforms
7) Speaker/headset driver for immediate playback
8 - 9) The Audio is processed in chunks that are (t) in length, usually in the order of 20-40ms
10) End user simultaneously providing input and receiving translated speech.

Figure 5: Real-time speech-to-speech pipeline: (a) wait-k listen/write policy and (b) corresponding signal path.

Direct end-to-end speech-to-speech models represent a relatively recent development within the broader family of large language and multimodal models. Unlike the cascading approach discussed in the previous section, these models do not rely on an explicit intermediate text representation; instead, linguistic processing is performed within a model trained directly on speech signals [SSJ+25]. This design can reduce end-to-end latency and mitigate error propagation that may arise from multiple processing stages in traditional multi-stage pipelines, and can support more natural conversational interaction with the system .

Within end-to-end speech-to-speech systems, two broad subcategories can be distinguished: direct models and real-time (streaming) models. Direct models process speech input as complete utterances, generating output only after the full input signal has been received. Their interaction pattern resembles cascading pipelines in terms of latency, as response generation is deferred until the entire input has been processed [ZFG+24].

In contrast, more recent real-time speech-to-speech systems, such as OpenAI's real-time speech model family adopt a streaming-based approach. Incoming speech is first digitized (e.g., using pulse-code modulation formats such as PCM16) and segmented into short audio frames on the order of tens of milliseconds. These segments are streamed incrementally to the model, which buffers a limited temporal context before beginning to generate spoken output [Ope24].

In Figure 5a, an example of such a system is illustrated using the simultaneous speech model introduced by Tan et al., showing how these models interleave listen and write operations according to a wait-k policy, allowing target-language output to begin before the full source utterance is observed [TLR+20]. To ground this abstract description in the physical signal chain used in safety-critical HMI deployments, Figure 5b annotates a representative streaming speech-to-speech path from the operator's microphone through synthesis and playback.

This streaming formulation enables incremental processing and response generation, which can reduce perceived latency and support a more natural conversational flow compared to full utterance-level processing approaches. However, a downside of such an approach is that end-to-end speech-to-speech models typically require larger computational resources compared with cascaded systems discussed previously, and they can face challenges in generalizing across diverse domains due to model architecture. Additionally, as of late 2025, native retrieval-augmented generation (RAG) compatibility remains an active area of research for end-to-end speech models, with most practical RAG implementations still relying on intermediate text representations for effective document retrieval [MML+25].

2.3 Vision-Language Models

Vision is another input channel that is of central importance to human psychological perception and interpretation, particularly in tasks where spatial reasoning and situational awareness are critical. Vision–language models extend machine interaction capabilities by enabling the joint processing of visual inputs and natural language, thereby allowing contextual understanding and reasoning grounded in visual information. These models exist at the intersection between computer vision and natural language processing [LWD+25]. For example, such models can be applied to the analysis of human facial expressions to infer affective states from visual data. An illustrative example of this type of application is provided in an accompanying notebook.[1]

[1] https://github.com/EmreMutlu99/Large-Language-Models-in-Human-machine-interaction/blob/main/tests/visual-language-models.ipynb

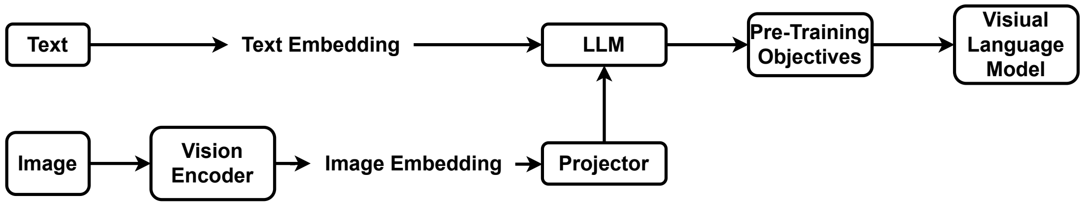


Figure 6: Conceptual VLM workflow: the vision encoder produces embeddings that are aligned and fused with LLM tokens to support multimodal reasoning and grounded responses.[LWD+25]

Vision-Language Models (VLMs) work by using a vision encoder to convert images into numerical feature representations and a large language model (LLM) to convert text into contextual embeddings. The vision encoder extracts visual features, which are then transformed into a compatible format before being integrated with the language model's input. In many modern VLMs, a pre-trained LLM serves as the backbone, and a projection or fusion layer is used to inject visual embeddings into the LLM so it can process visual information alongside text. This enables multimodal understanding and generation. A representation of this process is provided in Figure 6 [LWD+25]. In the context of human–machine interaction (HMI), such models can be deployed alongside audio- and text-based models to enhance the overall user experience in interactive systems.

3 Domain Case Studies

In the previous section, the architectural foundations underlying the use of large language models (LLMs) in the context of human–machine interaction were introduced, encompassing text-based, audio-based, and vision–language modalities. This section builds on that foundation by reviewing selected studies from several safety-critical application domains; namely aerospace and remote sciences, emergency and environmental response, and industrial operations, with a focus on how LLM-based technologies are applied in human–machine interaction within these contexts.

Each study is presented using a consistent structure, beginning with a brief overview of the application context, followed by a summary of the proposed method and key results. From the selected domains, one paper is classified as a design and interaction study, four papers focus on system and architecture design, and two papers are general survey studies.

3.1 Aerospace and Remote Science

Aerospace and remote science missions place humans in some of the most demanding HMI settings. Astronauts and mission operators work under high risk, limited communication, and long periods of isolation, while managing complex systems and strict safety procedures. In these conditions, interaction approaches that reduce cognitive effort, support decision-making, and offer context-aware assistance are especially valuable. In this section, selected papers from this domain are analyzed and discussed.

Conversational User Interfaces to Support Astronauts [FSG+21]

This paper uses a Wizard of Oz (WoZ) study[2] to investigate how astronauts would realistically interact with a conversational user interface (CUI) inside an extraterrestrial habitat before such a system technically exists. It investigates how CUIs can support astronauts living and working in extraterrestrial habitats (Moon/Mars), where isolation, confinement, delayed communication with Earth, and psychological stress are major challenges.

To address these differing interaction requirements, the authors designed two complementary CUIs: CASSIOPEIA, which provides task-oriented, mission-critical assistance for scientific operations, and PEGASUS, which functions as an emotionally expressive personal companion to support crew well-being.

While designing the conversational user interfaces (CUIs), the authors identified four key design guidelines. First, CUIs should support both mission-critical and non-mission-critical interactions, ensuring reliable task execution while also accommodating informal and supportive exchanges. Second, the interfaces should be designed for frequent and prolonged use, taking into account long-duration missions and sustained human–AI interaction without causing cognitive fatigue. Third, CUIs should enable multimodal and bidirectional interaction, allowing users to communicate with the system through complementary input and output modalities such as text, speech, and visual feedback. Finally, the CUI should be designed as a team member, positioned as an active collaborator that supports coordination, shared situational awareness, and trust within the crew.

Method

To derive requirements for conversational user interfaces in extraterrestrial habitats, the authors conducted a week-long Wizard-of-Oz study inside the MaMBA Moon/Mars habitat laboratory module at ZARM. Four experienced scientists carried out real biological, geological, and materials science experiments while interacting with a simulated CUI operated by a human wizard via a Raspberry Pi–based voice interface. All interactions were logged, video-recorded, transcribed, and qualitatively coded, revealing frequent use of both mission-critical and non-mission-related interactions. Based on these findings, the authors derived four design guidelines emphasizing support for mission and non-mission tasks, intensive long-term use, multimodal and bidirectional interaction, and the treatment of the CUI as a team member. Guided by these principles, the two CUIs introduced earlier were implemented. These systems were evaluated in two controlled user studies within the habitat: one comparing voice-only versus voice-plus-GUI interaction for task efficiency, and another comparing emotionally characterized versus neutral CUIs to assess emotional engagement, stress, and workload using standardized questionnaires and interviews.

[2] A Wizard of Oz study is an experimental method in which users interact with an apparently autonomous system whose behavior is secretly controlled by a human operator, which enables the study of interaction patterns before full system implementation.[DJA93]

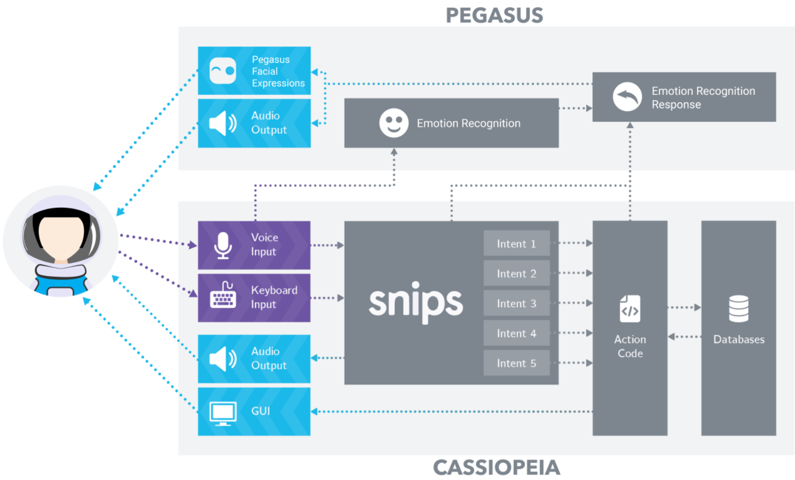


Figure 7: Overview of the PEGASUS and CASSIOPEI companion CUIs: multimodal astronaut inputs are parsed through intent recognition, emotional context, and action code to provide both operational and affective support.[FSG+21]

Result

The study shows that CUI can meaningfully support astronauts in isolated extraterrestrial habitats by improving both task performance and user experience. Results indicate that multimodal CUIs combining voice and graphical interfaces enable significantly more efficient task completion than voice-only CUIs, while perceived usability remains comparable. Emotionally characterized CUIs increase user excitement and engagement, and stronger emotional connection and perceived sympathy toward the system are associated with lower reported stress levels, although individual responses vary and overall task workload does not differ significantly. Overall, the findings suggest that well-designed CUIs can enhance operational efficiency and psychological well-being in long-duration space missions, provided that emotional interaction and interface modality are carefully adapted to user needs.

AI Assistants for Spaceflight Procedures: Combining Generative Pre-Trained Transformer and Retrieval-Augmented Generation on Knowledge Graphs With Augmented Reality Cues [BBN+24]

In this paper Bensch et.al. intoduce an intelligent personal assistant (IPA) they named CORE (Checklist Organizer for Research and Exploration). CORE is designed to support astronauts during procedural tasks on the ISS, Lunar Gateway, and future deep-space missions. It combines Large Language Models (GPTs) with Retrieval-Augmented Generation (RAG), Knowledge Graphs (KGs), and Augmented Reality (AR) to achieve its tasks.

Method

The authors validate CORE through qualitative, prototype-based functional testing rather than formal benchmarking. Ten publicly available ISS procedures were indexed into a vector database and linked with metadata and images via a knowledge graph, forming the basis of a retrieval-augmented generation (RAG) pipeline built on GPT-4. Spoken user queries were transcribed using Whisper, enriched with retrieved procedure content, and used to generate constrained responses. Validation was performed using example queries to verify that CORE returned the correct procedure steps verbatim and triggered associated visual cues, while rejecting non-procedure-related questions. In addition to this functional validation, offline availability is addressed as a system design consideration: the authors argue that recent advances in open-source large language models, together with optimization techniques such as low-rank adaptation and quantization, make fully offline deployment of the proposed architecture feasible for future missions. However, this offline capability is not empirically evaluated in the present work. No quantitative performance metrics or user studies are reported; a controlled evaluation at the European Astronaut Centre is explicitly deferred to future work.

Result

The authors demonstrate that the CORE prototype can correctly retrieve and present procedural information using a Retrieval-Augmented Generation architecture. In their experiments, CORE successfully returned exact, verbatim procedure steps in response to user queries, including the correct step numbering and associated visual references when available. The system was able to answer both step-specific questions (e.g., requesting a particular step in a CPR procedure) and metadata-related queries (such as the last update date of a procedure), reformulating retrieved graph data into coherent natural-language responses. Importantly, CORE consistently rejected questions unrelated to the indexed procedures, enforcing strict domain boundaries and reducing the risk of hallucinated or unsafe responses. These results indicate that the proposed combination of GPT, RAG, and knowledge graphs can support reliable, constrained procedural assistance. However, the reported results are illustrative and qualitative, and no quantitative performance metrics, comparative baselines, or user-study outcomes are provided.

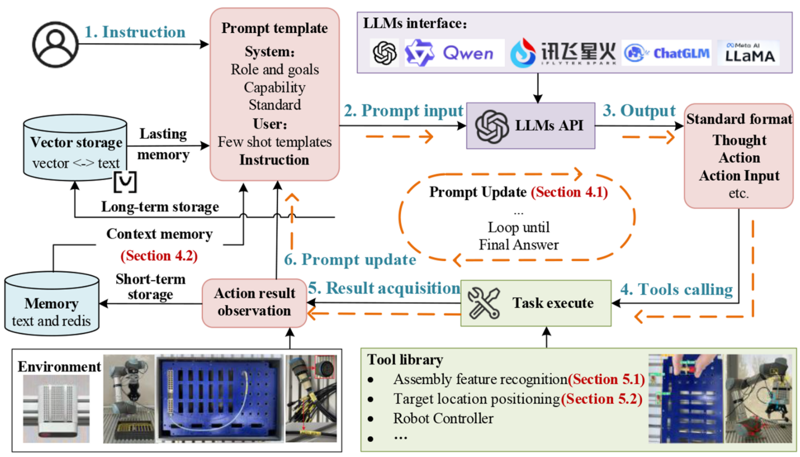


Figure 8: LLM-driven human–robot collaboration workflow showing how instructions propagate through prompting, memory, tool calls, execution, and feedback loops to support industrial assembly tasks. [WGZ+25]

LLM-Based Autonomous Agent for Aerospace Wire Harnessing [WGZ+25]

In this paper, Wang et al. introduce an agent-based framework for human–robot collaboration (HRC) and validate it through a high-precision aerospace wire harnessing assembly use case. Building on existing tool-calling and agentic reasoning paradigms discussed in Section 2, the framework extends these approaches by incorporating long-term memory mechanisms, implemented via vector database embeddings, to support contextual understanding, and iterative task execution.

The proposed system is an agent-based human–robot collaboration (HRC) framework in which a large language model (LLM) is embedded into a robotic system as an autonomous decision-making agent. Rather than treating the LLM as a passive instruction parser, the framework elevates it to an agentic role, capable of reasoning, planning, executing actions through tools, observing outcomes, and adapting behavior over time. Human operators interact with the system primarily through natural language instructions, provided in text or speech form, while the robot executes physical actions and provides implicit feedback through its movements and task outcomes; continuous collaboration is achieved by allowing the human to issue incremental or corrective instructions during execution, which are interpreted by the agent using contextual memory and integrated into subsequent planning steps.

The architecture consists of several interconnected modules that together enable task-oriented autonomous behavior. The configuration module defines the agent's role, goals, behavioral constraints, available tools, and prompt structure, ensuring that the LLM operates as a task-oriented autonomous agent rather than a conversational chatbot. Task planning is handled by a dedicated module that uses an unmodified LLM within a Reason–Act chain-of-thought loop to interpret human instructions, decompose complex tasks, and determine which actions and tools should be invoked at each step. The task execution module then carries out the planned actions by invoking encapsulated tools for perception, coordinate transformation, and robot control, and feeds the execution outcomes back to the agent as observations. Finally, a memory module maintains both short-term contextual memory and long-term vector-embedded summaries of past interactions, enabling continuous and context-aware reasoning across multiple tasks.

Method

The proposed HRC agent is validated experimentally using a real-world human–robot collaborative assembly setup based on a UR5 collaborative robot equipped with a depth camera and vision system. Validation is conducted progressively across three capability levels: perception, task planning, and task execution. First, the agent's sensing ability is evaluated using a Target Recognition Success Rate (TRR), which measures the correctness of object detection and spatial localization for tools, connectors, cables, and assembly ports under repeated trials, including dense connector arrays handled via a grid-matching recovery method. Second, the cognitive task-planning capability is assessed independently of robot execution by providing natural-language instructions and evaluating whether the agent's chain-of-thought reasoning produces correct and executable action sequences, quantified using the Task Planning Success Rate (TPR) and benchmarked against state-of-the-art standalone LLMs under identical tool constraints. Third, full end-to-end task execution is validated by deploying the planned action sequences on the physical robot and measuring the Task Execution Success Rate (TER), requiring all subtasks to be successfully completed in real assembly scenarios. Each task is repeated multiple times to assess robustness, and additional experiments demonstrate the agent's generalizability to previously unseen wire assembly tasks, thereby validating perception, reasoning, and execution capabilities in an integrated industrial setting

Result

The proposed HRC agent demonstrates consistently high success rates across perception, task planning, and execution in real aerospace wire harnessing assembly experiments, with most tasks achieving 9–10 successful trials out of 10 repetitions. In task planning, the agent substantially outperforms standalone state-of-the-art LLMs, which frequently fail to generate correct and executable action sequences for domain-specific assembly tasks. Overall, the results indicate that the agent-based LLM framework enables robust and reliable human–robot collaboration in complex industrial environments

3.2 Emergency and Environmental Response

Large Language Models (LLMs) offer new opportunities to support emergency and environmental response by enhancing situational awareness and decision-making under time pressure. In this section, a general review of their use in this field is presented.

Large Language Model Applications in Disaster Management: An Interdisciplinary Review [XMLC25]

Xu et al. provide a systematic interdisciplinary review of how large language models (LLMs) are being applied across the disaster management domain, taking into account both technical foundations and practical implementations. The paper analyzes 70 research studies published between 2020 and 2024 to chart the evolving role of LLMs in disaster risk reduction, emergency response, and resilience-building efforts.

The review characterizes large language models as human centered intermediaries that are capable of processing vast amounts of heterogeneous, unstructured, and time critical information. This includes data sources such as social media content, emergency situation reports, sensor streams, and remote sensing observations. Within disaster management settings, these capabilities enable LLMs to support human decision makers operating under uncertainty, high time pressure, and rapidly evolving conditions. The authors identify three recurring limitations in current disaster management systems. First, existing solutions tend to focus predominantly on the response phase, while offering comparatively limited support for preparedness, mitigation, and recovery activities. Second, there is insufficient integration among diverse stakeholders and heterogeneous data sources, which restricts the formation of a shared and coherent operational picture. Third, although situational awareness is often achieved, it is frequently not translated effectively into concrete and actionable intelligence that can directly inform decisions. To address these challenges, the paper introduces the *3M Framework* as a conceptual foundation for next generation LLM driven disaster information management systems.

Method

The authors carried out a systematic literature review guided by the PRISMA[3] methodology, drawing on major academic and preprint databases including Web of Science, IEEE Xplore, ScienceDirect, Scopus, and arXiv. Following a two stage screening process, a total of 70 studies were selected for inclusion. These studies employ large language models or contextual language models such as GPT, BERT, and T5 within disaster related applications. The selected works were examined along two primary dimensions. The first dimension concerns the operational phase of disaster management addressed by each study, including detection, tracking, analysis, and action.

[3]The PRISMA methodology (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) provides evidence-based guidelines for transparently reporting systematic reviews by defining a structured process for study identification, screening, and inclusion. [PMB+21]

The second dimension focuses on the technical approach, covering aspects such as the underlying model architecture, the chosen adaptation strategy, whether fine tuning or prompt engineering, and the degree of integration with external tools or additional data sources. This structured synthesis allows for a comparative perspective on how LLM capabilities are applied and evolve across different disaster types and across the various phases of disaster management.

Result

Large language models show strong performance in detecting disaster-related content, including misinformation filtering, humanitarian needs classification, and severity estimation from informal and multilingual data. Multimodal models further improve early warning by jointly processing text, images, and sensor data. For tracking, LLMs enhance event monitoring through entity extraction, temporal reasoning, and spatial interpretation; zero-shot methods reduce reliance on annotated datasets, though fine-grained disambiguation across locations and entities remains challenging. In analysis, LLMs support impact assessment via sentiment modeling, topic evolution, semantic networks, and knowledge graphs, enabling evaluation of response effectiveness and stakeholder communication. In the action phase, LLMs operate as interactive planning assistants, coordinating responders and affected populations through retrieval-augmented generation, tool orchestration, and scenario analysis.

Decoder-focused models, such as the GPT-4 family, increasingly dominate due to strong reasoning and generative capabilities, while encoder-based models remain important for low-latency and resource-constrained settings.

To guide future research and deployment, the authors propose the 3M Framework: multi-modal data fusion, integrating textual, visual, sensor, and structured data into unified situational assessments; multi-source information validation, cross-checking heterogeneous and conflicting sources to improve reliability under uncertainty; and multi-agent knowledge integration, coordinating human experts, AI agents, and physical–virtual systems to produce actionable intelligence.

From Promising Capability to Pervasive Bias: Assessing Large Language Models for Emergency Department Triage [LSB$^+$25]

Lee et al. investigate the applicability of large language models (LLMs) for emergency department (ED) triage as high-level cognitive collaborators for professionals. The study is motivated by the limitations of both human triage, which is subject to cognitive bias and variability under pressure, and traditional machine learning approaches, which rely heavily on structured inputs and often fail under distribution shifts.

The authors position LLMs as decision support tools rather than autonomous decision-makers, capable of processing unstructured clinical text and generating triage predictions and rationales.

Method

The study employs a two-part experimental design. First, the authors benchmark LLMs for emergency department triage, evaluating robustness to missing data and distribution shifts across three datasets: MIMIC-IV-ED hospital records, KTAS-labeled hospital data, and curated ESI handbook cases. These datasets combine structured clinical variables with free-text chief complaints. LLM-based approaches, including zero-shot, chain-of-thought, few-shot prompting, retrieval-based in-context learning (KATE), and supervised fine-tuning, are compared against traditional machine learning baselines and a BioBERT-based model using accuracy, F1-score, quadratic-weighted kappa, and mean squared error. Second, a counterfactual analysis systematically perturbs patient sex and race to assess demographic biases in LLM triage predictions using statistical significance testing.

Result

The results show that LLM-based approaches outperform traditional machine learning baselines across most datasets, particularly when retrieval-based in-context learning is used. The KATE method achieves the strongest performance, emphasizing the importance of retrieving relevant examples, while fine-tuned smaller models remain competitive with sufficient domain data. However, notable limitations are identified. LLMs show calibration issues by hesitating to assign the highest acuity level, and the counterfactual analysis reveals significant demographic biases across intersections of sex and race. Although chain-of-thought prompting reduces some bias, it does not eliminate it.

3.3 Industrial Operations

Industrial operations represent an early proving ground for LLM-enabled human–robot collaboration workflows, where operator intent can be translated into contextual tool use, perception, and execution. This section presents a paper that explores the potential application of LLM-based technologies in the context of human–machine interaction (HMI).

Human–robot collaborative visual inspection with Large Language Models [Tas26]

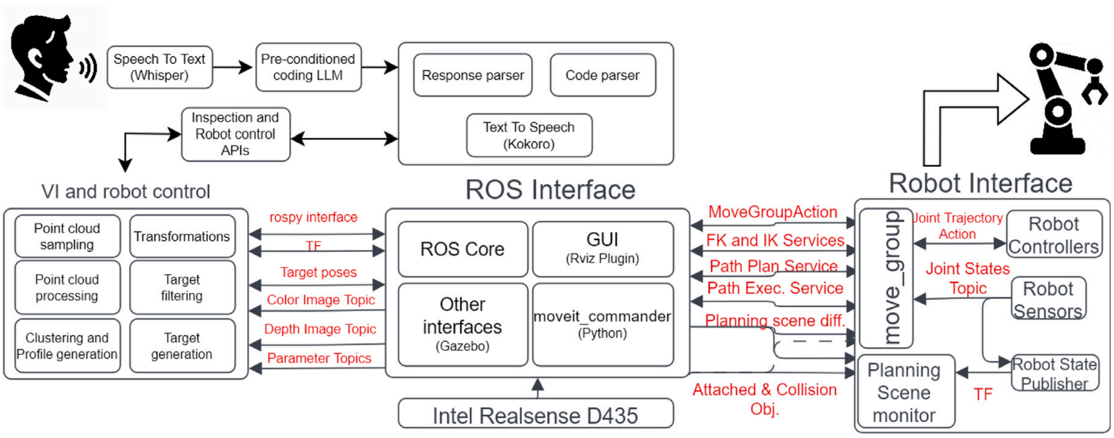


Figure 9: Pipeline of the human–robot collaborative inspection system: speech input is parsed by LLM-based components that interface with ROS and deterministic robot controllers for safe, repeatable execution.

Tasneem et al. present a human–robot collaborative (HRC) visual inspection system that integrates large language models (LLMs) to enable intuitive, speech-based interaction between human operators and industrial robots. The work is motivated by the limitations of traditional GUI- and teach-pendant-based robot programming, which can be cognitively demanding, inflexible, and costly in environments that require frequent reconfiguration. The system uses LLMs to translate natural language instructions into robot inspection actions via deterministic local control. It supports human inspectors by reducing cognitive load and aiding decision-making in complex Industry 5.0 inspection tasks.

Method

As shown in Figure 9, the system adopts a hybrid control architecture that separates high-level reasoning from low-level execution. Spoken commands are transcribed using a neural speech-to-text model and processed by a locally deployed, code-generating LLM, which produces task-specific code based on structured prompts defining task constraints and robot APIs. High-level planning, including inspection path generation and task sequencing, is handled by the LLM, while low-level motion control, kinematics, and collision avoidance are executed deterministically using ROS and MoveIt. A closed-loop text-to-speech interface provides execution feedback to the user.

To ensure suitability for industrial deployment, the authors benchmark multiple local LLMs using criteria including prompt adherence, code correctness, inference latency, and computational footprint, selecting a model that balances execution reliability with real-time performance under local hardware constraints.

Result

Experiments in simulated and real industrial settings show that the system performs complex visual inspection tasks across objects with varying sizes, geometries, and clutter. The selected lightweight, code-oriented LLM balances prompt adherence, execution reliability, and computational cost, enabling real-time operation.

The hybrid architecture supports robust inspection path generation, collision avoidance, and multi-object inspection while maintaining operator control and safety. It improves interaction ergonomics and reduces reliance on expert robot programming. Limitations include speech recognition errors in noisy environments and performance trade-offs related to model size. Overall, the results demonstrate that combining agentic LLM-based planning with classical robotic control is effective for human-centered industrial automation.

3.4 Discussion

As discussed in Section 2, large language models are currently employed in three main interaction modalities, namely text based, audio based, and vision based models, as well as in combined multimodal configurations. Across all reviewed domains, the literature indicates that these technologies are not deployed as standalone decision making systems, but rather as extensions of traditional human machine interaction mechanisms. In this role, LLM based interfaces often function as alternative interaction media, effectively supplementing or partially replacing conventional graphical user interfaces and control elements such as buttons and menus. In addition, several of the reviewed works employ LLMs as semi autonomous assistants that support human operators through information retrieval, interpretation, and task coordination, while remaining embedded within constrained and human supervised system architectures.

Despite these advantages, the direct use of LLMs in safety critical contexts introduces fundamental challenges, including hallucinations and limited reliability. The reviewed works therefore employ a range of architectural and interaction techniques introduced in Section 2, such as retrieval augmented generation and agentic control structures, in order to enhance performance and mitigate these limitations. In this context, two central research questions guide the discussion.

- *(RQ1)* How do these techniques strengthen the existing advantages of large language models when applied to safety critical human machine interaction?
- *(RQ2)* To what extent do they address challenges intrinsic to large language model based interaction, particularly with respect to reliability, hallucinations, and operator trust?

In aerospace applications, these questions are reflected in conversational user interface designs that emphasize long term interaction, multimodal communication, and psychological support under conditions of isolation and high cognitive load. Freitas et al. [FSG+21] demonstrate that LLM driven conversational interfaces can serve both mission critical and non mission critical roles, supporting procedural tasks while also contributing to crew well being. From the perspective of RQ1, this illustrates how natural language interaction and multimodal feedback can improve usability and reduce cognitive burden and help create a CUI as an alternativ for a GUI. Similarly, the procedural assistant proposed by Bensch et al. [BBN+24] shows how text based and audio based interaction pipelines can be combined with retrieval augmented generation to deliver strictly bounded, procedure specific assistance. This architecture directly targets hallucination reduction and reliability, aligning with RQ2, while reinforcing the role of the LLM as a mediating layer between human intent and verified system knowledge rather than as an autonomous controller.

In industrial and robotic settings, LLM based interaction mechanisms are increasingly coupled with agentic planning and tool calling architectures, while low level execution remains delegated to deterministic control programs. The human robot collaboration systems presented by Wang et al. [WGZ+25] and Tasneem et al. [Tas26] exemplify this approach by using LLMs for task interpretation, planning, and coordination, while relying on classical robotic control stacks such as ROS and MoveIt for safe and repeatable execution. These systems highlight how agentic reasoning and natural language interfaces can be used as an interface with Robotics, addressing RQ1, while strict separation between high level reasoning and low level control mitigates safety and reliability risks, addressing RQ2. Although vision based and multimodal inputs are incorporated to support perception and inspection, linguistic reasoning remains the dominant modality for decision making and coordination.

In emergency and healthcare related contexts, LLMs primarily function as high level cognitive aids that process unstructured textual information to support time critical decision making. The interdisciplinary survey by Xu et al. [XMLC25] shows that text based LLMs are widely applied across detection, tracking, analysis, and action phases of disaster management, where they enhance situational awareness and coordination without replacing human judgment. These findings illustrate how LLMs can amplify existing human capabilities in information processing and sense making, addressing RQ1. In a clinical setting, Lee et al. [LSB+25] demonstrate that LLMs can outperform traditional machine learning approaches in emergency department triage by effectively handling free text inputs, while also revealing important limitations related to calibration and demographic bias. These results directly address RQ2, highlighting that technical performance gains must be accompanied by careful consideration of trust, fairness, and human oversight when integrating LLM based interfaces into safety critical workflows.

Taken together, the reviewed literature suggests a convergence toward hybrid interaction architectures in which LLMs operate as cognitive intermediaries that augment human capabilities across text, audio, and visual modalities. While the specific interaction modality and degree of autonomy vary by domain, the underlying design principle remains consistent. LLMs are most effective when used to enhance interpretability, coordination, and usability, while safety, accountability, and final authority are preserved through deterministic execution layers and human in the loop control. At the same time, the psychological and social dimensions of human machine interaction, such as trust, perceived reliability, and user comfort, emerge as critical factors that shape both the effectiveness and acceptance of LLM based systems, pointing toward important limitations and directions for future research.

4 Conclusion

This work reviewed recent applications of large language models in safety-critical human–machine interaction, encompassing text-based, audio-based, and visual language interfaces. Across the examined domains, LLMs demonstrate the greatest value when deployed as assistive systems that support interpretation, planning, and decision-making rather than as autonomous controllers. Studies in aerospace, emergency response, and industrial settings consistently highlight the effectiveness of hybrid architectures that separate high-level reasoning from deterministic low-level execution.

Overall, the findings indicate that successful integration of LLMs, including speech-enabled and vision-language models, depends on constrained agentic designs, human-in-the-loop control, and clear system boundaries. While challenges related to bias, calibration, multimodal integration, and reliability remain, the evidence suggests that carefully designed multimodal LLM-based interfaces can reduce cognitive load and improve usability without compromising safety. Future work should focus on retrieval-grounded reasoning, multimodal fusion, and human-centered evaluation to support deployment in real-world safety-critical environments.

Bibliography

[BBN+24] Oliver Bensch, Leonie Bensch, Tommy Nilsson, Florian Saling, Bernd Bewer, Sophie Jentzsch, Tobias Hecking, and J. Nathan Kutz. Ai assistants for spaceflight procedures: Combining generative pre-trained transformer and retrieval-augmented generation on knowledge graphs with augmented reality cues, 2024. arXiv:2409.14206 [cs.AI].

[BKK+24] Vineet Bhat, Ali Umut Kaypak, Prashanth Krishnamurthy, Ramesh Karri, and Farshad Khorrami. Grounding llms for robot task planning using closed-loop state feedback. *arXiv preprint arXiv:2402.08546*, 2024. Planning prompt format and BodyLLM vs. embedding matching comparison discussed on p. 6.

[BMR+20] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020.

[CFD24] Kevin Carolan, Luke Fennelly, and Mark Doyle. A review of multi-modal large language and vision models. *arXiv preprint arXiv:2404.01322*, 2024. Survey of transformer-based multimodal LLMs integrating vision, audio, and language.

[CWF23] Sarah D. Childress, Tara C. Williams, and David R. Francisco. Nasa space flight human-system standard: enabling human spaceflight missions by supporting astronaut health, safety, and performance. *npj Microgravity*, 9:31, 2023.

[CYJ+25] Wei Cui, Dong Yu, Xiaoqiang Jiao, Zhehuai Meng, Guanyu Zhang, Qiang Wang, Yike Guo, and Irwin King. Recent advances in speech language models: A survey. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 2025. Survey of transformer-based speech language models, speech-to-speech systems, and extensions of text-based LLMs to speech.

[DFAB04] Alan Dix, Janet Finlay, Gregory D. Abowd, and Russell Beale. *Human–Computer Interaction*. Pearson Education, Harlow, England, 3 edition, 2004. pp. 38–54 summarize human sensory channels, memory systems, reasoning, and emotional factors relevant to HCI design.

[DJA93] Nils Dahlbäck, Arne Jönsson, and Lars Ahrenberg. Wizard of oz studies — why and how. *Knowledge-Based Systems*, 6(4):258–266, 1993.

[FSG+21] Ana Rita Gonçalves Freitas, Alexander Schülke, Simon Glaser, Pitt Michelmann, Thanh Nguyen Chi, Lisa Schröder, Zahra Fadavi, Gaurav Talekar, Jette Ternieten, Akash Trivedi, Jana Wahls, Warda Masood, Christiane Heinicke, and Johannes Schöning. Conversational user interfaces to support astronauts in extraterrestrial habitats. In *Proceedings of the 20th International Conference on Mobile and Ubiquitous Multimedia (MUM '21)*, Leuven, Belgium, 2021. ACM. Wizard-of-Oz analog mission deriving CUI design guidelines for lunar/martian habitats.

[GYA+24] Zorik Gekhman, Gal Yona, Roee Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. Does fine-tuning llms on new knowledge encourage hallucinations? *arXiv preprint arXiv:2405.05904*, 2024. See p. 9 for related-work discussion of new knowledge and hallucinations.

[HUWN+] Reinhold Haeb-Umbach, Shinji Watanabe, Tomohiro Nakatani, Michiel Bacchiani, Bjoern Hoffmeister, Michael L. Seltzer, Heiga Zen, and Mehrez Souden. Speech processing for digital home assistants. Draft manuscript prepared for IEEE Signal Processing Magazine; cited content on p. 7 (Fig. 3).

[Int16] International Electrotechnical Commission, Geneva, Switzerland. *IEC 60204-1:2016 – Safety of machinery – Electrical equipment of machines – Part 1: General requirements*, 2016. Edition 6.0.

[JLF+24] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Delong Chen, Wenliang Dai, Ho Shu Chan, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *arXiv preprint arXiv:2202.03629*, 2024. p. 39.

[KLM24] Callie Y. Kim, Christine P. Lee, and Bilge Mutlu. Understanding large-language model (llm)-powered human-robot interaction. In *Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*, Boulder, CO, USA, 2024. ACM. User study (pp. 2–6) comparing LLM-powered robots against text/voice agents across choose/generate/execute/negotiate tasks.

[LSB+25] Joseph Lee, Tianqi Shang, Jae Young Baik, Duy Duong-Tran, Shu Yang, Lingyao Li, and Li Shen. From promising capability to pervasive bias: Assessing large language models for emergency department triage. *arXiv preprint arXiv:2504.16273*, 2025. Version 2.

[LWD+25] Zongxia Li, Xiyang Wu, Hongyang Du, Fuxiao Liu, Huy Nghiem, and Guangyao Shi. A survey of state of the art large vision language models: Alignment, benchmark, evaluations and challenges. *arXiv preprint arXiv:2501.02189*, 2025. pp. 1–10.

[MML+25] Do June Min, Karel Mundnich, Andy Lapastora, Erfan Soltanmohammadi, Srikanth Ronanki, and Kyu Han. Speech retrieval-augmented generation without automatic speech recognition. *arXiv preprint arXiv:2412.16500*, 2025. Accepted at ICASSP 2025; content referenced from p. 1.

[Mos23] Marius Mosbach. Analyzing pre-trained and fine-tuned language models. Lecture notes, Saarland University Big Picture seminar, 2023. Accessed 2024.

[OBME24] Oded Ovadia, Menachem Brief, Moshik Mishaeli, and Oren Elisha. Fine-tuning or retrieval? comparing knowledge injection in llms. *arXiv preprint arXiv:2312.05934*, 2024. p. 8 covers the fine-tuning vs. RAG comparison conclusions.

[Ope24] OpenAI. Realtime api documentation. `https://platform.openai.com/docs/guides/realtime`, 2024. Accessed: 2025-12-15.

[PMB+21] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, et al. The prisma 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021.

[RSS+25] C. Rathan, S. Sagar, R. K. Shreekrishana, S. K. Sudeep, and M. Govinda Raju. Human machine interface (hmi) – a survey. *International Journal of Innovative Research in Technology*, 11(11):e174698, 2025. Department of Electronics and Communication Engineering, RV College of Engineering; referenced via research/IJIRT174698_PAPER.pdf.

[SDYD+23] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, Roberta Raileanu, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761*, 2023. pp. 3–4 describe Toolformer's tool-sampling/filtering pipeline.

[SSJ+25] Mohammad Sarim, Saim Shakeel, Laeeba Javed, Jamaluddin, and Mohammad Nadeem. Direct speech to speech translation: A review. *arXiv preprint arXiv:2503.04799*, 2025. Overview of cascade vs. direct S2ST trade-offs; cited pp. 1–2.

[Tas26] Osama Tasneem. Human–robot collaborative visual inspection with large language models. *Robotics and Computer-Integrated Manufacturing*, 98:103154, 2026. Early view article (2025).

[TLR+20] Xu Tan, Jinglin Liu, Yi Ren, Chen Zhang, Tao Qin, Zhou Zhao, and Tie-Yan Liu. Simulspeech: End-to-end simultaneous speech to text translation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3787–3796. Association for Computational Linguistics, 2020.

[VBBK23] Sai Vemprala, Rogerio Bonatti, Arthur Bucker, and Ashish Kapoor. Chatgpt for robotics: Design principles and model abilities. *arXiv preprint arXiv:2306.17582*, 2023. pp. 2–6 outline the function-library pipeline, prompt engineering tactics, and dialog-driven corrections.

[VSP+17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *arXiv preprint arXiv:1706.03762*, 2017.

[WGZ+25] Yiwei Wang, Qi Guo, Lianyu Zheng, Binbin Wang, Pai Zheng, and Zhonghua Qi. Llm based autonomous agent of human-robot collaboration for aerospace wire harnessing assembly. *Robotics and Computer-Integrated Manufacturing*, 97:103120, 2025. Available online 2 September 2025.

[XMLC25] Fengyi Xu, Jun Ma, Nan Li, and Jack C. P. Cheng. Large language model applications in disaster management: An interdisciplinary review. *International Journal of Disaster Risk Reduction*, 127:105642, 2025.

[YZY+23] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2023. ICLR 2023 paper introducing the ReAct prompting paradigm.

[ZFG+24] Shaolei Zhang, Qingkai Fang, Shoutao Guo, Zhengrui Ma, Min Zhang, and Yang Feng. Streamspeech: Simultaneous speech-to-speech translation with multi-task learning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1–23, 2024.